

Variance and Correlation Analysis Report

Pipeline outputs are generated from the selected input table.

The report includes data-cleaning notes, group summaries, ANOVA effect sizes, and a correlation matrix.

Source: D:\\\\u5de5\\u4f5c\\u6d41\\portfolio\\projects\\stats_variance_correlation_pipeline\\data\\example_measurements.csv

Rows: original=60, cleaned=60

Removed: missing=0, duplicates=0

Groups: Control, Treatment_A, Treatment_B

Metrics: nematode_density, plant_height, root_biomass, soil_nitrate

Group summary

metric	group	n	mean	sd	variance	median	se
nematode_density	Control	5	40.8	2.387	5.7	41	1.068
nematode_density	Treatment_A	5	30	1.581	2.5	30	0.7071
nematode_density	Treatment_B	5	23	1.581	2.5	23	0.7071
plant_height	Control	5	18.16	0.493	0.243	18.2	0.2205
plant_height	Treatment_A	5	21.7	0.5874	0.345	21.7	0.2627
plant_height	Treatment_B	5	24.78	0.6221	0.387	24.8	0.2782
root_biomass	Control	5	2.1	0.1581	0.025	2.1	0.07071
root_biomass	Treatment_A	5	2.9	0.1581	0.025	2.9	0.07071
root_biomass	Treatment_B	5	3.58	0.1924	0.037	3.6	0.08602
soil_nitrate	Control	5	12.12	0.4438	0.197	12.1	0.1985
soil_nitrate	Treatment_A	5	16.42	0.4658	0.217	16.3	0.2083
soil_nitrate	Treatment_B	5	19.7	0.5874	0.345	19.7	0.2627

One-way ANOVA results

metric	groups	n	df_between	df_within	f_statistic	p_value	eta_squared
nematode_density	3	15	2	12	112.7	1.666e-08	0.9495
plant_height	3	15	2	12	168.8	1.634e-09	0.9657
root_biomass	3	15	2	12	94.62	4.496e-08	0.9404
soil_nitrate	3	15	2	12	285.6	7.591e-11	0.9794

Correlation matrix

metric	nematode_density	plant_height	root_biomass	soil_nitrate
nematode_density	1	-0.9857	-0.9893	-0.9857
plant_height	-0.9857	1	0.9964	0.9821
root_biomass	-0.9893	0.9964	1	0.9857
soil_nitrate	-0.9857	0.9821	0.9857	1

group_mean_se.png





